

```
#!/bin/bash
network_id="xxxxxxxxxxxxxx"
# Check if already connected to the
network
if sudo zerotier-cli listnetworks | grep
-q "$network_id"; then
    echo "Already connected to network
    $network_id"
else
    # Join the ZeroTier network
    echo "Joining ZeroTier network
    $network_id..."
    join_output=$(sudo zerotier-cli join
    $network_id)

    if [[ "$join_output" == *"200 join
    OK"* ]]; then
        echo "Successfully joined network
        $network_id"
    else
        echo "Failed to join network.
        Output: $join_output"
    fi
fi

# Get ZeroTier info and list networks
sudo zerotier-cli info
sudo zerotier-cli listnetworks
sudo zerotier-cli info
```

Create a systemd service file to run the script at boot:

```
[Unit]
Description=Run ZeroTier Script at Boot
After=network.target

[Service]
ExecStart=/path/to/your/zerotier.sh
ExecReload=/bin/kill -HUP $MAINPID
Restart=on-failure
User=pi # Or your username

[Install]
WantedBy=multi-user.target
```

\$ > sudo nano /etc/systemd/system/zerotier.service

Activate and test the service with the following commands:

TO DOWNLOAD SOURCE CODE AND RESOURCES



Scan this QR code or go to <https://sourcecode.electronicshub.in> to access and download source code and other resources for this project

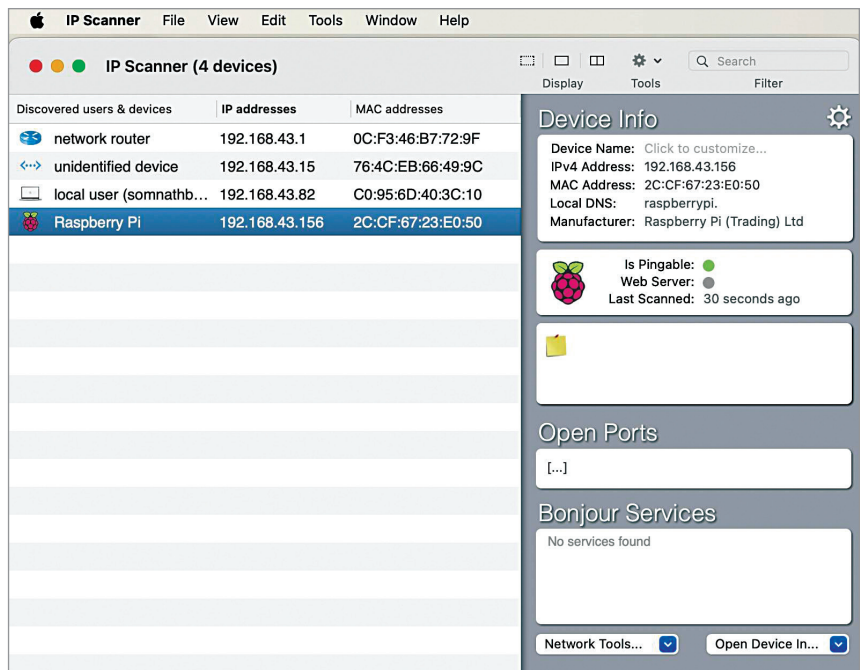


Fig. 6: Raspberry Pi Zero with ZeroTier setup

```
$> sudo systemctl daemon-reload # Reload
systemd to recognize the new service
$> sudo systemctl enable zerotier.service
# Enable the service to run at boot
$> sudo systemctl start zerotier.service
# Start the service immediately
$> sudo systemctl status zerotier.service
# Check the service status (optional)
```

This setup ensures that the Pi Zero checks for the ZeroTier network on every boot and joins it if not already connected.

IP addresses on Wi-Fi LAN

Modern mobile hotspots often mask the IP addresses assigned to connected devices, making it challenging to identify the Raspberry Pi's IP, particularly for SSH access.

To simplify this, install a network scanning tool such as LanScan (macOS), advanced IP scanner (Windows), or nmap (cross-platform) on a laptop or desktop. These tools quickly display all devices and their IPs on the Wi-Fi network, making it easier to locate and access the Pi.

Aftermath

This system demonstrates the versatility of the Raspberry Pi Zero as a compact, low-power solution for live

streaming, remote monitoring, and IoT applications, without reliance on high-power or resource-intensive hardware. With efficient energy consumption, the Pi Zero operates reliably for long durations, even in mobile or battery-powered setups, distinguishing it from full-sized Raspberry Pi boards.

By using open source tools such as Motion for video streaming and ZeroTier for secure remote access, the system provides a stable, self-contained solution for one-to-one streaming and long-distance monitoring, entirely free of proprietary platforms or costly subscriptions.

Looking ahead, the Raspberry Pi Zero has the potential to evolve into a more powerful platform by integrating AI-based image analysis, real-time alerts, and edge computing. These enhancements could expand its applications into security, environmental monitoring, and remote data acquisition, making it a future-ready solution for dynamic, real-time scenarios. EFY



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